

# Hierarchical maximum Margin Preservation under Bounded Score Perturbations

Research Architecture Runtime  
operator/hierarchical-maximum

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## Abstract

Hierarchical maximum reduces to unique-maximizer Argmax margin stability:  $\gamma > 2\varepsilon$  preserves the winner under  $\|\delta\|_\infty \leq \varepsilon$ . Lean: `LEAN.FULL`.

## 1 Theorems

**Theorem 1** (Preservation). *Definitional reduction to Argmax margin invariance.*

**Theorem 2** (Sharpness). *Definitional reduction to Argmax margin sharpness.*

## 2 Lean / certificate

`Research.Operators.HierarchicalMaximum.Margin.`

## 3 Limitations

Mathlib  $\mathbb{R}$  (`REAL_MATHLIB`); Argmax reduction.